Geological Everything Model 3D: Unified Interactive Inference through Target-Conditioned Representations

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ABSTRACT

Scientific imaging requires the inference of spatial objects, geometric boundaries, and continuous properties from limited observations. Some tasks in medical imaging, remote sensing, and subsurface imaging share the challenges of missing observations and target ambiguity: high-dimensional data and large-scale training do not necessarily supply all the information needed to determine a target. Local annotations, domain interpretations, and independent measurements can provide additional target constraints. An important question is how these heterogeneous conditions can form representations jointly with observations, enabling one model to infer different targets and revise its predictions. We propose Geological Everything Model 3D (GEM), which explores unified three-dimensional interactive inference through target-conditioned representations. Object membership, geometric relationships, and measured values are encoded as local conditions and supplied with observations to a three-dimensional backbone, allowing conditions to participate in feature extraction from the input stage. Multitask joint training brings discrete object recognition and continuous field estimation into a common conditional prediction interface. An interactive workflow allows conditions to be accumulated, revised, and revisited. Adding, replacing, or withdrawing conditions updates candidate results while retaining valid historical constraints, enabling continued inference with fixed model parameters. Subsurface imaging serves as the principal validation setting, where complex three-dimensional structures, diverse targets, and sparse constraints provide a challenging test. Synthetic benchmarks, field surveys, and blind-well evaluations demonstrate applications to fault and horizon interpretation, geobody identification, stratigraphic coordinates, and physical property estimation. GEM outperforms the compared baselines on field fault slices and some tasks constrained by well logs, whereas specialized models retain advantages on some synthetic tasks, indicating that the benefits depend on the setting. A Martian radar example provides exploratory evidence for application across imaging modalities. These results support the feasibility of the approach for subsurface interpretation and modeling. This work provides an approach to unified modeling and interactive revision for heterogeneous target inference under insufficient observations, with potential extensions to medical imaging and other remote-sensing tasks.

Keywords: target-conditioned representations; unified three-dimensional inference; human-in-the-loop interpretation; subsurface imaging; geological modeling.
Project website: `https://douyimin.github.io/GEM/`

1 Introduction

Interpretation and modeling in scientific imaging require spatial objects, geometric boundaries, and continuous properties to be inferred from limited observations. High dimensionality and large data volumes do not imply sufficient target information: different target states may produce similar responses, whereas the same target may appear differently under different imaging conditions. We refer to this insufficiency relative to target discrimination and estimation as low effective information density. The term describes how strongly observations constrain a target, rather than the statistical entropy of the raw data. Poor imaging quality can reduce this information, but even clearly resolved image features may leave their target identity or physical meaning ambiguous.

Figure 1a illustrates the problem through progressively reduced visual cues. Color helps separate neighboring objects; grayscale removes this cue; a gradient-dominated representation suppresses much of the surface texture and

intensity context; and an outline representation further discards continuous gradient-strength information. As fewer cues remain, similar boundaries can admit more competing assignments of object identity and extent, making the depicted instances increasingly difficult for an AI model or a human observer to distinguish. Local annotations, domain interpretations, and independent measurements can supply information that the remaining image does not determine. The central question is how such conditions can participate in representation formation together with observations to support inference of different targets and allow predictions to change as conditions are revised.

This problem occurs in a range of tasks in medical imaging, remote sensing, and subsurface imaging. In medical ultrasound, speckle, acoustic shadows, and missing signals make tissue boundaries difficult to identify completely, requiring anatomical priors and spatial context[1]. In synthetic aperture radar, layover, shadow, and speckle make image appearance depend jointly on viewing geometry and scattering mechanisms, increasing ambiguity in land-surface interpretation[2]. Remote-sensing inversion of vegetation parameters is limited by parameter coupling and nonuniqueness, requiring priors and auxiliary observations to constrain continuous properties[3]. These settings differ in imaging physics and target meaning but share insufficient observational constraints on the target. More training data and larger models can improve learned priors without necessarily supplying target information missing from a particular scene. Predictions may also become unreliable when the statistical associations on which a model relies change between settings. It is therefore necessary to investigate how local conditions can be translated into target constraints that a model can use.

Local conditions can specify object membership and geometric position or anchor continuous values. Although their forms differ, they all constrain the current prediction target at a subset of spatial locations. Object recognition, boundary delineation, and continuous field estimation may also use common relationships, including spatial adjacency, interfaces, and regional correspondence, providing a basis for investigating shared representations. These commonalities motivate a unified conditional prediction interface while retaining the physical meaning, numerical scale, and error requirements of each task.

The SAM family shares a common approach: reusable visual representations support segmentation guided by prompts that specify a target, with interactive revision of predictions[4–6]. This paradigm has extended to medical imaging and subsurface interpretation[7,8], providing a basis for studying how heterogeneous conditions can jointly support inference of three-dimensional objects and continuous fields.

Local target conditions can enter three-dimensional feature extraction alongside observations. They can supply object membership or measured values absent from the image and direct the model toward existing cues relevant to the target. Information absent from both observations and conditions remains a limit to inference.

Conditions also change during interpretation. After a model produces a three-dimensional candidate result, an expert can use domain knowledge and local observations to annotate missed structures, correct boundaries, or exclude errors, prompting the model to update its prediction. Conditions must therefore be accumulated, replaced, and withdrawn, preserving valid earlier judgments while allowing a return to previous states. Human interpretations, independent measurements, and inherited predictions should retain their respective provenance. Interaction thus becomes a process of continuing revision of conditional predictions, allowing experts to adjust target constraints and compare interpretations.

We investigate whether jointly representing observations and local conditions allows one model to infer discrete objects and continuous fields when image information is insufficient. The conditions specify the current target and can be revised through expert feedback, allowing interpretation to proceed as geological understanding develops.

We evaluate GEM primarily on subsurface images, where complex three-dimensional structures must be interpreted from sparse labels and measurements. Structural interpretation and property modeling also underpin resource assessment, subsurface storage, and geological hazard research[9,10]. Figure 1b contrasts an outcrop, where material appearance and layer contacts can be observed directly, with a seismic section, where geological structures must be inferred from indirect wave responses. Seismic imaging does not preserve a direct view of the subsurface: wave propagation, finite acquisition, limited bandwidth, incomplete illumination, and processing restrict which geological contrasts become visible. Multiple subsurface models can therefore remain compatible with the same observations[11]. Faults may resemble dipping horizons, and channel flanks or basal boundaries may lack continuously traceable reflections (Fig. 1c)[12]. Compared with directly visible geological features, identifying structures, correlating horizons, or inferring lithology from these responses depends more strongly on contextual knowledge. Where the observations cannot distinguish competing interpretations, expert judgments informed by regional geology and independent measurements can be essential for obtaining a usable result.

Separate models for faults, channels, relative geological time (RGT), and acoustic impedance can accommodate their respective targets and error requirements[13–16]. These targets may nevertheless use common stratigraphic correspondence, truncation, and spatial adjacency relationships. Their transfer difficulty differs: some fault detection

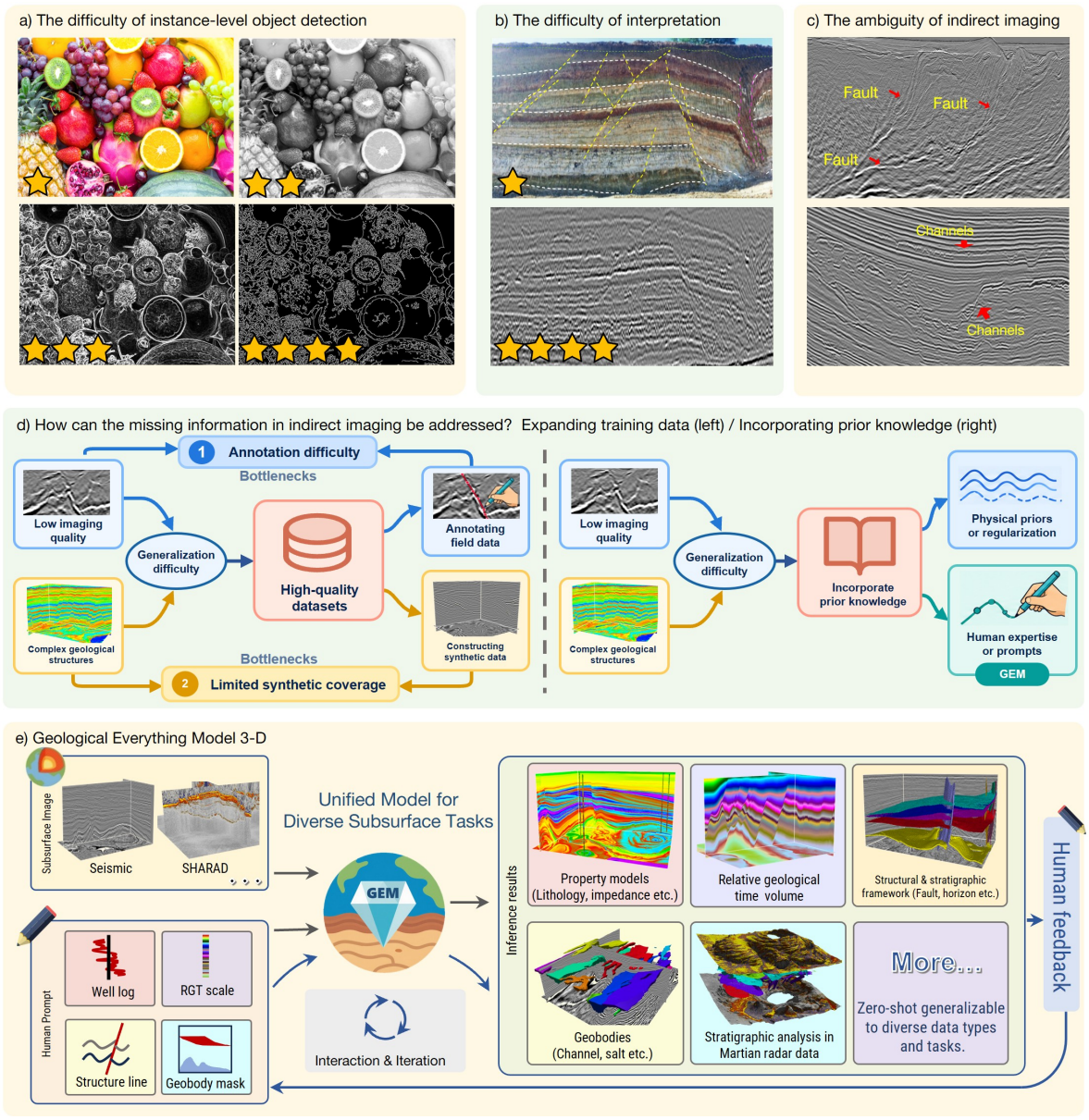


Figure 1. Information loss, interpretation ambiguity, and the GEM approach. (a) Color, grayscale, gradient-dominated, and outline representations illustrate progressively reduced cues for distinguishing object identity and extent. Removing color and texture context and then continuous gradient-strength information makes the depicted instances harder to distinguish. (b) Directly visible layer contacts and material appearance in an outcrop contrast with indirect geological responses in a seismic section; the images are not a registered pair. (c) Fault and channel examples show how similar or incomplete reflections leave target identity and boundaries ambiguous. (d) Two complementary routes: expanding training data to cover geological and imaging variability, limited by field-annotation difficulty and synthetic coverage; and supplying additional constraints through physical priors or human expertise. GEM develops the latter route through local interpretations and measurements. (e) Images and spatial conditions from well logs, RGT reference scales, interpretation lines, or masks jointly enter GEM. Human feedback revises the conditions, allowing predictions to be updated with the same model weights across surveys and tasks. Outputs depict separate applications, not a single joint prediction (Appendix C.1).

relies on relatively stable local displacement cues, whereas channels, karst cavities, and salt domes depend more strongly on object scale and geological context[12]. Complex geological assemblages vary across surveys, while changes in acquisition bandwidth, illumination, noise, and processing alter how those assemblages are expressed in images.

Together, geological variability and limited observational information weaken the stability of learned associations, so accuracy in a source survey does not necessarily imply reliable performance elsewhere[17]. Evaluation across tasks and surveys must therefore consider shared capabilities, task differences, and the role of conditional inputs together.

Figure 1d summarizes two complementary responses to this generalization problem. The first expands training data to cover a wider range of geological configurations and imaging conditions, allowing networks to learn statistical relationships between image responses and structures, stratigraphic organization, or property distributions. Synthetic data provide spatially aligned images and complete targets[18,19], but their coverage depends on the geological processes, geometries, and acquisition effects represented by the simulator. Complex combinations of deformation, deposition, and heterogeneity make exhaustive coverage difficult. Annotating field data introduces real geological variability, yet weak or incomplete reflections and nonunique interpretations make dense three-dimensional labels costly to construct and difficult to define consistently. These bottlenecks limit the range of conditions that a training dataset can represent.

Representation learning can improve the use of available data. The Seismic Foundation Model learns seismic representations through masking[20]; SparK provides masked pretraining for convolutional backbones[21]; and domain prompts and lightweight tuning further extend these applications[22]. Such approaches strengthen learned priors and adaptation, but increasing data coverage alone does not necessarily resolve which of several compatible targets is intended in a particular scene. A statistical relationship learned across examples cannot supply every local fact absent from the current observation.

The second response introduces additional constraints into prediction. Physical relationships and geological regularization restrict the range of admissible solutions, while expert interpretations and independent measurements provide information about the current target and setting[10,23]. These contributions are complementary: physical consistency constrains what can occur, and local knowledge helps specify which object, boundary, or property is sought. When additional observations are difficult to obtain and existing data cannot determine the target, translating domain-informed human interpretations into explicit constraints is one of the few practicable routes to improving imaging or interpretation results. GEM develops the local-information route in Fig. 1d,e by encoding interpretation geometry, object membership, and measured values as spatial conditions. It combines priors learned from training data with information supplied for the current application, enabling experts to refine target constraints through interaction.

We propose Geological Everything Model 3D (GEM) to explore unified three-dimensional interactive inference through target-conditioned representations. As illustrated in Fig. 1e, observed images and task-encoded local conditions enter a three-dimensional backbone together, allowing conditions to participate in feature formation from the input stage. In the subsurface implementation, masked seismic pretraining provides an observational representation, mixed multitask supervision learns mappings from local conditions to three-dimensional targets, and an independently trained structural supervision network provides learned geological constraints on dense outputs. The model unifies the conditional prediction interface for discrete objects and continuous fields while retaining the encoding and numerical processing required by each task. Complete condition states are managed for individual targets. Adding, replacing, and withdrawing conditions supports multiround inference with fixed parameters, while saved historical states allow backtracking and alternative interpretations.

We examine the method using synthetic benchmarks, field surveys, and a Martian radar example. Slice comparisons, full-field synthetic evaluation, and blind-well tests assess prediction for the corresponding tasks. RGT and radar examples illustrate the application range, while feature visualizations provide clues about representation mechanisms. The results support the use of one model for multiple conditioned subsurface targets. We describe the application protocol as fixed-weight conditional inference and distinguish transfer across surveys, target types, and task families in Appendix C.1. Similar information bottlenecks in medical imaging and remote sensing provide the broader motivation and suggest possible extensions to these domains.

The main contributions are:

- A target-conditioned three-dimensional inference framework for insufficient observations, implemented in a unified model, GEM. Object membership, geometric constraints, and numerical anchors are encoded as local conditions and supplied with observations to a three-dimensional backbone, so that conditions participate in feature extraction from the input stage. Masked pretraining, multitask joint training, and learned structural supervision support discrete object recognition and continuous field estimation in the same model.

- Three-dimensional conditional inference driven by local interpretation feedback. Expert review and revision of candidate results are translated into updatable spatial constraints. With fixed model parameters, observations and valid historical conditions jointly update predictions to alleviate ambiguity in target identity and spatial

extent, support local correction and comparison of alternative hypotheses, and provide a methodological basis for improving the reliability of complex subsurface interpretation.

- Application and evaluation across surveys, tasks, and imaging modalities. With fixed model weights and local conditions from each survey, GEM performs fault and horizon interpretation, geobody delineation, RGT estimation, and property modeling constrained by well logs. Slice comparisons, full-field synthetic evaluation, and blind-well tests assess prediction across surveys and tasks, while indirect imaging data from different modalities are used to explore the applicability of unified conditional inference.

2 Three-dimensional geological interpretation constrained by local conditions

2.1 Target-conditioned inference from indirect observations

GEM encodes local information (Fig. 1d,e) by converting interpretations and measurements into spatial conditions that enter the model together with an existing image volume. Observations provide spatial cues such as reflection continuity, displacement, truncation, and regional morphology, while local conditions supply the identity, position, or value of the current target. Together they define the information available for prediction, allowing expert understanding of subsurface structures and independent measurements to inform estimates in unspecified regions.

Conditions from different sources are represented by condition values $\mathbf{V}$ and a validity mask $\mathbf{M}$, spatially aligned with the observations. Interpretation lines specify interface locations, local masks identify object membership, RGT reference sections provide relative stratigraphic coordinates, and well logs supply property values at known locations. The validity mask distinguishes unlabeled locations from known background or zero values. Let $\mathbf{X}$ denote the observed image. The prediction is

$$\widehat{\mathbf{Y}} = G_\theta\big(\mathbf{X}, \mathcal{E}(\mathbf{M}, \mathbf{V})\big), \tag{1}$$

where $\mathcal{E}$ encodes conditions according to the target type and G_θ is the shared three-dimensional model. Geometry, categories, and continuous values are thus incorporated into a common conditional prediction formulation while retaining their geological meanings and numerical scales.

We use the term information injection for the participation of spatially encoded local knowledge and measurements in prediction. This supplies target information that observations alone cannot readily determine and guides the model toward target-relevant cues within those observations. For object interpretation, conditions specify the object of interest and its local extent; for property estimation, they anchor local values. The model combines these constraints with spatial priors learned during training to infer three-dimensional distributions. Target ambiguity in indirect observations is thereby formulated as a prediction problem that can be constrained by local information and revised as understanding develops.

2.2 Joint representation and learning of geological spatial relationships

Unified modeling is motivated by geological spatial relationships that can be shared across targets. Interface interpretation delineates boundaries, geobody identification defines regions, and property modeling estimates numerical distributions. All three can use spatial organization associated with stratigraphic distribution, interface contacts, and structural deformation. GEM therefore learns correspondences between local conditions and complete three-dimensional targets, enabling geometric delineation, region identification, and numerical estimation within a common representation.

Observations and spatially aligned conditions are fused at the input and jointly participate in three-dimensional feature extraction. Interface or object conditions guide the network to use reflection continuity, termination, and displacement to form the corresponding target representation. Well-log conditions provide numerical anchors, while structural cues in the images provide spatial context for interwell prediction. The influence of local conditions on unspecified regions is mediated by spatial relationships learned by the network. Property prediction uses the image volume and well-log conditions directly; geological structure participates through cues and latent representations, without first requiring a separate structural framework or segmentation result.

This correspondence from local information to three-dimensional targets is established through three complementary learning stages. Masked pretraining reconstructs missing seismic blocks to learn reflection organization and spatial context. Multitask joint training samples local conditions from spatially aligned structure, geobody, RGT, and property labels within the same geological setting, supervising outputs with complete targets. An independently trained and subsequently frozen structure-aware network provides auxiliary supervision for dense predictions, constraining geological spatial structure that can be assessed through fault, channel, and karst recognition. The

training workflow is shown in Supplementary Fig. S1; encoding, architecture, and optimization details are given in Appendix A.

All results across surveys and tasks in this study are obtained with the same model weights, sharing both the backbone and output head. Local conditions guide internal representations for predicting different targets. Continuous properties are restored to their respective numerical scales using the normalization parameters of the input well logs. This design incorporates geological context and local target information into inference, enabling one model to address different forms of observational insufficiency through a common information-injection mechanism.

2.3 Application across surveys and interactive revision

When applied to a new survey, the model receives the local image volume and interpretations or well logs while its parameters remain fixed. When the target changes, local conditions guide the formation of the corresponding internal representation, and the shared output head produces a prediction. Examples include changing from local interface conditions to object-mask conditions, or from relative stratigraphic coordinate conditions to well-log property conditions.

Interpreters can review three-dimensional results, annotate missed structures, modify existing interpretations, or withdraw incorrect conditions, then rerun prediction using all valid conditions. Conditions and their provenance are saved at each round to support backtracking and comparison of alternative interpretations (Supplementary Fig. S4). Measurements, human interpretations, and inherited predictions are recorded separately. Inherited predictions may retain earlier errors and cannot be treated as new independent information.

3 Results

The experiments test the same conditional inference mechanism under three forms of observational insufficiency: ambiguous interface identity, incomplete geobody boundaries, and sparse measurements of continuous properties. Structural interpretation tests how local geometric knowledge guides spatial continuation; geobody identification tests how local membership conditions specify objects with different morphologies; and property modeling tests how numerical measurements constrain fields between sampled locations. Synthetic references, field interpretations, and blind wells assess these predictions. All applications use the same model weights and shared output head, with the target specified through local conditions. Appendix C.1 defines the evaluation protocol.

3.1 Structural features and responses to conditions

Fault displacement, unconformity truncation, and stratigraphic termination all involve relationships between a target and surrounding reflections. Figure 2a juxtaposes expert interpretations and channel features from the model's first upsampling stage. The interpretations are used only for comparison and were not supplied to this feature-extraction process. After principal component analysis (PCA), the first three components are mapped to RGB space. Some projected boundaries coincide with interpreted structures, indicating that intermediate features contain information related to stratigraphic organization and structural discontinuities.

Figure 2b illustrates the joint constraints of well-log values and seismic structure on property prediction. In Fig. 2c, a lightweight multilayer perceptron (MLP) projects features from modules k_1 to k_5 while GEM parameters remain frozen. Under mask conditions, responses are concentrated around the specified object and its neighborhood; under well-log conditions, they exhibit a layered distribution. The modules k_1 to k_5 denote network depth, not interaction rounds. These projections describe the spatial distribution of feature responses under each condition type. Inference uses the deterministic mapping in Eq. (1); the probability notation in the figure denotes conditional dependence only. The different response patterns are consistent with local conditions directing shared features toward object extent or stratigraphic variation.

3.2 Structural and stratigraphic interpretation

Faulting, folding, and erosion cause displacement, termination, and local loss of seismic reflections. Low-angle thrust faults and detachment surfaces are particularly easy to confuse with horizons, and local reflection events alone may not determine an interface's geological meaning. GEM uses interpretation lines to specify an interface and combines them with surrounding reflections to predict its three-dimensional geometry.

Fault evaluation includes 100 synthetic volumes of 512^3 voxels and 20 expert-annotated slices from the Hikurangi and Baiyun surveys. Compared methods are U-Net, DeepLabV3, HRNet, FaultNet, and FaultSeg+[13, 24–28]. GEM achieves an IoU of 0.6700 on synthetic data, below HRNet's 0.6891, and 0.6268 on field slices, above the other plotted methods, among which HRNet reaches the highest score of 0.1583 (Fig. 3a). The comparison uses each method's adaptation workflow, as specified in Appendix C.

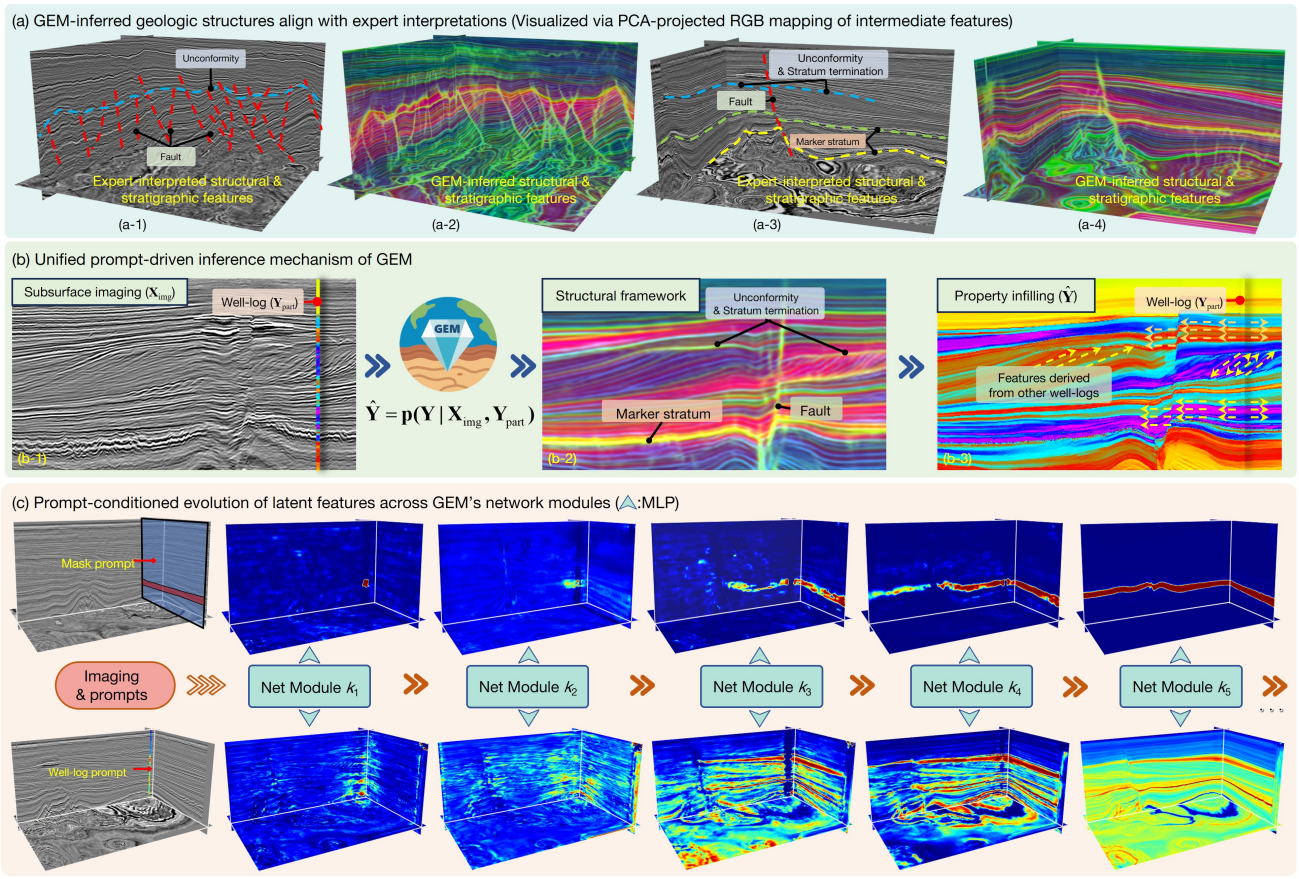


Figure 2. Structure-related representations and responses to local conditions. (a-1, a-3) Seismic images and expert interpretations; (a-2, a-4) PCA–RGB projections of the corresponding intermediate features. Expert interpretations are used to compare spatial correspondence and were not inputs to this feature-extraction process. (b) Conceptual relationship among observations, well-log conditions, and property predictions. Probability notation denotes conditional dependence; the implemented model is a deterministic predictor. (c) MLP feature projections for network modules k_1 to k_5 under mask and well-log conditions. Colors indicate projected responses, not confidence.

The Hikurangi subduction zone provides a field setting with pronounced target ambiguity. Basement relief associated with subducting seamounts, thrust structures, and sediment deformation produce complex reflection assemblages. Distinguishing low-angle faults from horizons requires spatial relationships and regional geological knowledge[29]. In Fig. 3b,c, interpreters mark targets on local sections, and GEM extends them into three-dimensional surfaces. Examples include the Pāpaku fault, the Tuaheni Ridge fault, other thrust faults, detachment surfaces, and a bottom-simulating reflector (BSR). Different conditions specify different interpreted objects; reflection interfaces such as the BSR are therefore handled separately from fault surfaces. The illustrated instances use an average of 2.4 prompts.

The same model is applied to Baiyun using local seismic data and interpretation lines. Supplementary comparisons for the two surveys show differences between GEM and fault-specific methods in target coverage and surface geometry (Supplementary Fig. S6). These surfaces can provide geometric boundaries for mechanical models of subduction zones, supporting tests of how fault geometry and its configuration relative to sediment bodies influence shallow slip. Fluid and slow-slip mechanisms still require independent observations[29].

For Martian SHARAD radar imaging, the same model extracts reflection interfaces using local interpretation lines. Figure 3d includes the surface, four internal interfaces, and the basal interface, with an average of 5.8 prompts per instance. This example explores reflection imaging beyond seismic data. Although the imaging physics differ, interface tracking in both modalities is affected by weak signals and locally missing reflections. Background on SHARAD three-dimensional imaging is provided in Refs.[30,31].

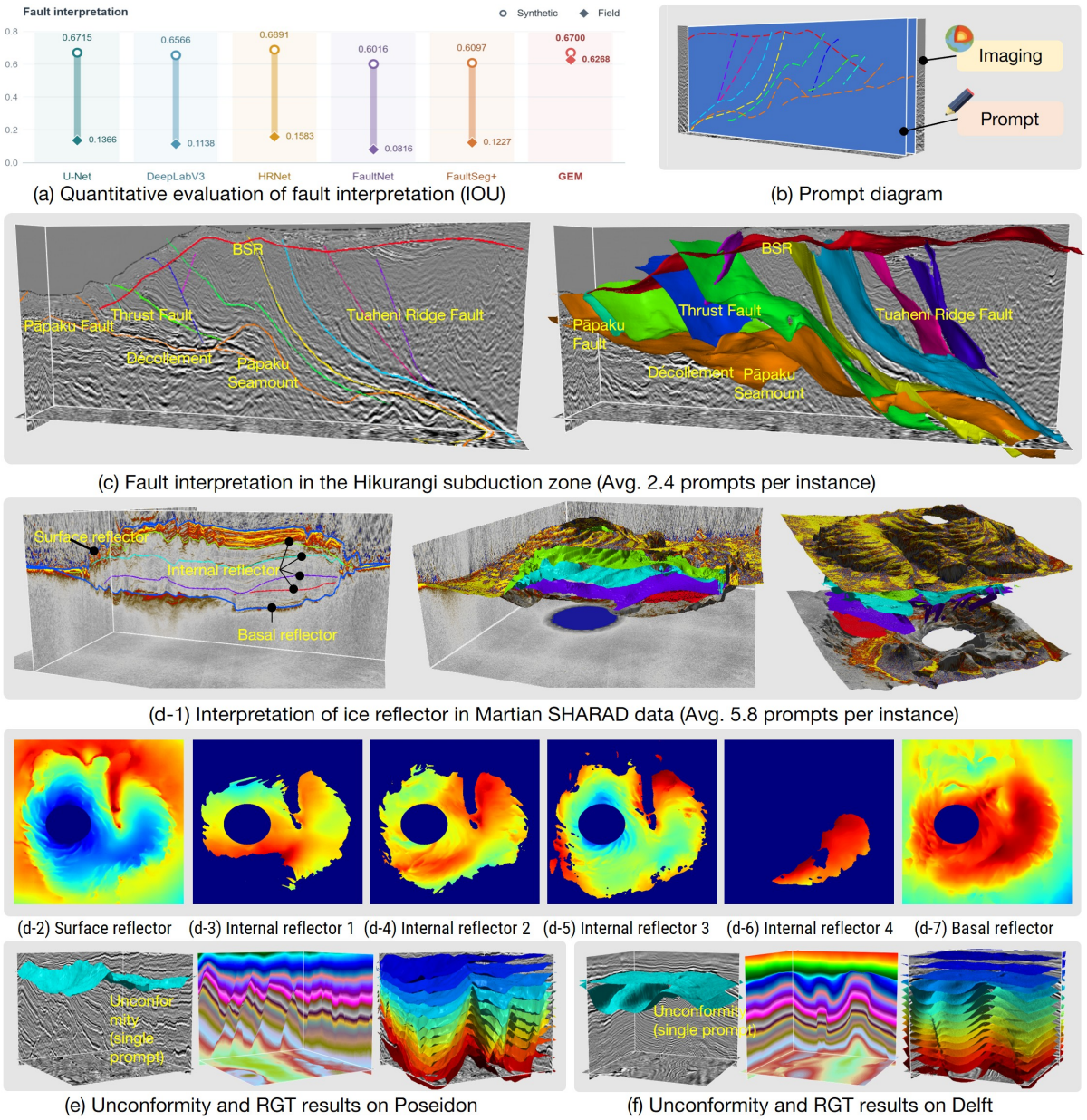


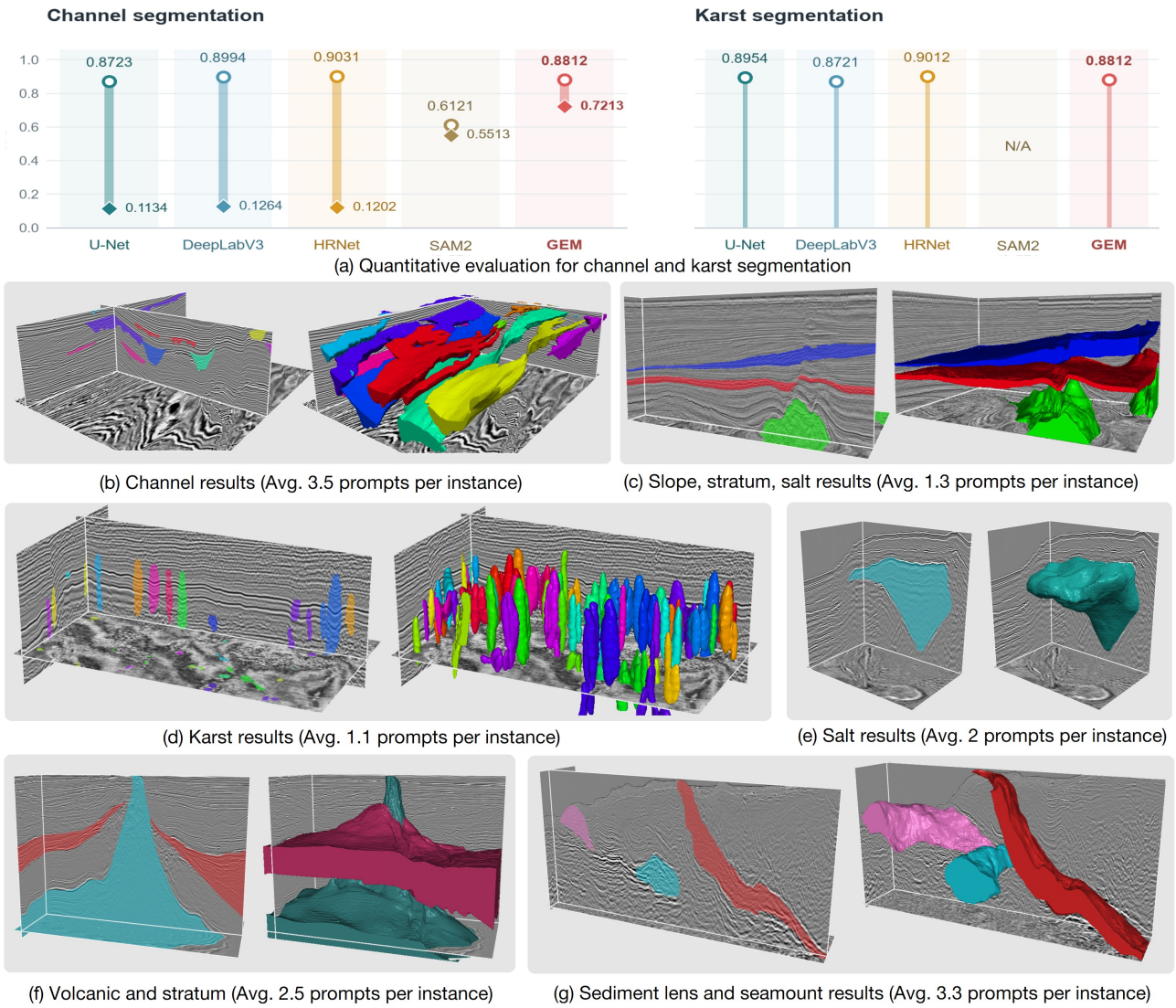
Figure 3. Reconstruction of structures, reflection interfaces, and stratigraphic coordinates from local interpretations. (a) Fault IoU: open circles and filled diamonds denote synthetic and field data, respectively. Lines connect two scores for the same method and are not error intervals. (b) Imaging data and local interpretation-line inputs. (c) Local interpretations and three-dimensional surfaces at Hikurangi, with an average of 2.4 prompts per instance. (d-1) SHARAD reflection-interface interpretation, averaging 5.8 prompts per instance; (d-2)–(d-7) show the surface, four internal interfaces, and the basal interface. (e, f) Unconformities, RGT, and stratigraphic isosurfaces at Poseidon and Delft; the single-prompt annotation applies to the unconformity instances. Colors in (c, d-1) distinguish interpreted objects; RGT color scales in (e, f) denote relative stratigraphic coordinates. Neither represents confidence.

The Poseidon and Delft data contain faults, unconformities, and reflection terminations. In addition to local interfaces, GEM estimates RGT fields from reference sections to describe relative stratigraphic position within the volume (Fig. 3e,f). Within a survey, unconformity interpretation uses local geometric conditions, whereas RGT estimation uses relative-coordinate conditions, whose isosurfaces describe three-dimensional stratigraphy. The

single-prompt annotation in the figure applies only to unconformity instances; RGT uses iterative self-conditioning (Supplementary Fig. S5). RGT is a relative stratigraphic coordinate, not an absolute depositional age.

Across these surveys, local interpretation lines select interfaces whose geological identity is ambiguous in the image, while reference coordinates guide RGT estimation. The same model extends both forms of local information through the reflection volume. These results support conditional inference for recovering spatial continuity from incomplete observations, with the radar example extending interface extraction to another imaging modality.

3.3 Three-dimensional geobody identification



(a) Quantitative evaluation for channel and karst segmentation

(b) Channel results (Avg. 3.5 prompts per instance)

(c) Slope, stratum, salt results (Avg. 1.3 prompts per instance)

(d) Karst results (Avg. 1.1 prompts per instance)

(e) Salt results (Avg. 2 prompts per instance)

(f) Volcanic and stratum (Avg. 2.5 prompts per instance)

(g) Sediment lens and seamount results (Avg. 3.3 prompts per instance)

Figure 4. Geobody selection and three-dimensional delineation using local masks. (a) Channel and karst segmentation IoU: open circles denote synthetic data, and filled diamonds denote field channels. Karst comparisons include synthetic data only; N/A marks entries without a plotted score. (b) Parihaka channels. (c) Slope, stratigraphic, and salt bodies in the Netherlands F3 survey. (d) Fort Worth karst bodies. (e) Gulf Coast salt bodies. (f) Kora volcanic units and strata. (g) Hikurangi sediment lenses and a seamount. Each group shows local conditions and three-dimensional results, with colors distinguishing instances or geological units. Prompt counts describe the interaction workflow and do not imply equal annotation amounts or human time.

Channels, karst, salt bodies, and volcanic units differ substantially in morphology and reflection character. Targets within the same category also vary with depth, depositional setting, and imaging conditions. Local masks specify locations belonging to the target and regions to exclude, allowing GEM to estimate its three-dimensional

volume using surrounding reflections. In contrast to interface tracking, the output here is an object mask, with backbone parameters unchanged.

Channel testing includes 100 synthetic samples and 30 annotated field slices from Parihaka and Romney; karst testing includes 20 synthetic samples[18,19]. On synthetic channels and karst, GEM achieves an IoU of 0.8812 for both, whereas HRNet reaches 0.9031 and 0.9012, respectively. On field channels, GEM achieves 0.7213, compared with 0.5513 for SAM 2 and 0.1134–0.1264 for the other plotted methods (Fig. 4a). As in fault evaluation, GEM performs better relative to the compared methods on the annotated field data than on the synthetic benchmark.

Parihaka channels exhibit scale variation, branching, and incomplete boundaries; local reflection discontinuities may also result from faults. Local masks distinguish the target channel from similar reflections. The predicted volumes reveal its sinuous course and branches, with an average of 3.5 prompts per instance (Fig. 4b). The same workflow is applied to Romney, where supplementary comparisons show differences in cross-section continuity relative to a slice-based extension of SAM 2 (Supplementary Fig. S7).

Figure 4 also includes Fort Worth karst bodies; slope, stratigraphic, and salt bodies in the Netherlands F3 survey; Gulf Coast salt bodies; Kora volcanic units and strata; and Hikurangi sediment lenses and a seamount. The respective mean prompt counts per instance are 1.1, 1.3, 2, 2.5, and 3.3. These targets range from elongated branches and local cavities to large geological units, all specified through local masks. These examples extend object segmentation to different geological categories using the same weights.

At Hikurangi, extracted fault surfaces, sediment lenses, and the seamount provide a basis for comparing spatial relationships between structures and sediment bodies. Objects are predicted from independent conditions, without explicit geological relationship constraints between objects.

Branching channels, localized cavities, salt bodies, and sedimentary units differ markedly in shape and scale, yet local membership masks allow the same model to delineate them. Together with the interface results, these applications show that specifying target identity and local extent enables one model to reconstruct both surfaces and volumes from indirect observations.

3.4 Property modeling constrained by well logs

The limited bandwidth of seismic responses restricts recovery of absolute property trends and fine-scale variations. Well logs provide local values at relatively high vertical resolution but do not directly determine interwell variation. GEM uses well logs as numerical conditions and seismic data as three-dimensional spatial context to estimate velocity, acoustic impedance, and natural gamma ray separately. Each property is processed according to its units and numerical range, while lithology uses a categorical representation.

SEAM I provides a complete velocity model for evaluating full-field prediction from sparse conditions. The model contains salt-related structures, strong velocity contrasts, and lateral variation. With four well-like sections as conditions, GEM predictions capture the main salt geometry and large-scale velocity trends (Fig. 5b). Its MAE is 0.1085, a 27.47% reduction from the best baseline, DeepLabV3, at 0.1496. SSIM is 0.8119, a 4.29% increase over HRNet's 0.7785, and FID is 107.1, a 2.19% reduction from DeepLabV3's 109.5 (Fig. 5a). MAE evaluates numerical deviations; SSIM and FID provide complementary comparisons of local structure and appearance distributions, respectively[32,33]. Comparison methods are trained using the same four sections, whereas GEM receives them as conditions during inference; the evaluation therefore compares two adaptation workflows.

Field property evaluation uses blind wells excluded from the conditioning set. The Teapot Dome example shows an acoustic impedance field and two blind-well curves (Fig. 5c). The aggregate MAE across four evaluation wells is 123.9, slightly below HRNet's 125.6, corresponding to a 1.35% improvement.

The Netherlands F3 example uses three conditioning wells and holds out F02-1 as a blind well. Predicted acoustic impedance retains the main layered variations and structural boundaries (Fig. 5d). At F02-1, GEM's MAE is 252.8, compared with 295.2, 429.2, and 496.1 for HRNet, DeepLabV3, and U-Net, respectively, a 14.36% reduction relative to the best baseline. On the same seismic volume, local conditions for the corresponding targets also produce natural gamma-ray and lithology predictions without refitting the network. The log curves and lithology strips show agreement and local discrepancies for these additional targets.

Delft acoustic impedance and natural gamma-ray predictions are compared with blind wells PNA-13 and PNA-GT-01 in Fig. 5e. Predictions capture major vertical trends but differ locally from measurements. Poseidon sonic transit time and natural gamma-ray results are shown in Supplementary Fig. S8. Across these surveys, the image and local measurements change while the model weights remain fixed.

These property results extend the information-injection mechanism from specifying geometry and object membership to constraining numerical values. The synthetic field evaluation and predictions at blind wells support its use for estimating properties beyond the supplied measurements, with improvements that vary across surveys.

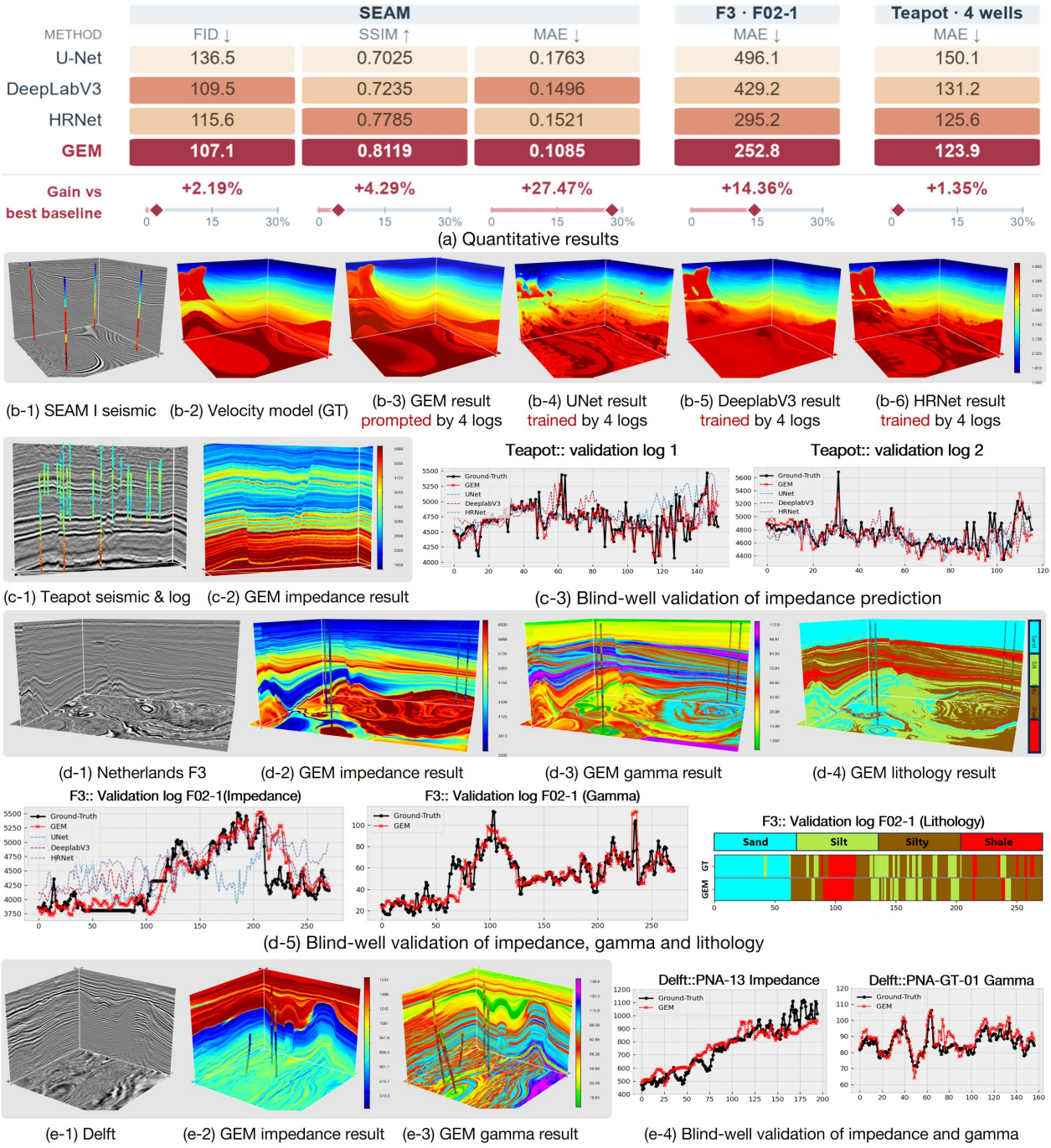


Figure 5. Property modeling with sparse well constraints. (a) Metrics for SEAM, F3 F02-1, and four Teapot wells. Cells show original values; dark red, salmon, pale peach, and pale beige indicate ranks 1–4 within each metric. Lower FID and MAE and higher SSIM are better. The lower row shows GEM's improvement over the best baseline for each metric. Across the five columns, these baselines are DeepLabV3, HRNet, DeepLabV3, HRNet, and HRNet. Improvements use $100(b-g)/b$ for FID/MAE and $100(g-b)/b$ for SSIM, where b and g are baseline and GEM values; all axes span 0–30%. (b) SEAM I seismic data, true velocity, and four predictions. GEM uses four well-like conditions; comparison methods are trained on the same sections. (c) Teapot acoustic impedance and blind-well curves. (d) F3 acoustic impedance, natural gamma ray, lithology, and the F02-1 blind-well comparison. (e) Delft property fields and blind-well comparisons. MAE values for different targets are not directly compared across datasets.

Together with the structural and geobody results, these predictions show that local conditions can specify geometric, categorical, or continuous targets while the same model uses image structure to estimate their spatial distribution. Blind-well coverage is limited, and multiple compatible interwell solutions may remain. Differences in units and normalization also preclude direct comparison of MAE across datasets.

4 Discussion

4.1 The role of local conditions in subsurface interpretation

Indirect imaging records the response of the subsurface to probing signals. Structures in an image do not always correspond uniquely to geological objects or properties. Local interpretations and well logs provide two different kinds of information: interpreters specify target identity and extent, whereas well logs supply local numerical values. The former depend on geological understanding, and the latter are affected by measurement and well-to-seismic registration errors. GEM uses these sources together with three-dimensional observations, allowing different targets to be interpreted from the same image volume under different conditions.

In many indirect imaging settings, missing observations are accompanied by incomplete understanding of target identity, spatial relationships, and physical meaning. When several interpretations are compatible with the observations and the contextual knowledge needed to distinguish them is inadequately represented in the data or training examples, increasing model size alone may not determine the intended target. Human judgments based on regional geology, independent observations, and research objectives may then be necessary to obtain usable imaging or interpretation results. GEM incorporates such judgments into existing image volumes as local spatial conditions. Interpreters can then compare the resulting three-dimensional predictions with geological evidence and revise the conditions. Expert knowledge becomes an explicit, revisable part of the inference process.

GEM outperforms the compared methods in field fault and channel evaluations, whereas specialized models achieve higher IoU on some synthetic segmentation tasks. Property improvements also vary: SEAM velocity MAE decreases by 27.47% relative to the best baseline, but Teapot acoustic impedance MAE decreases by only 1.35%. The effectiveness of local conditional prediction therefore depends on the data, target, and supplied conditions. Some baselines use those conditions during training, whereas GEM uses them at inference.

4.2 Scope of the shared model

Structure, geobody, and property prediction share a conditional input format and backbone parameters while retaining their geometric, categorical, and numerical output requirements. Stratigraphic correspondence, truncation, and adjacency may provide common cues for several targets, and Fig. 2 shows structure-related responses. Feature projections, however, describe spatial correspondence only.

Across surveys, local conditions specify the target without changing the model weights. The information supplied by an interpretation line differs from that in a full-section mask or an inherited prediction. Prompt count alone therefore cannot measure annotation effort; the spatial extent of the conditions and the time spent by the interpreter also matter.

4.3 Interpretation revision and remaining uncertainty

Interaction allows interpreters to modify or withdraw conditions and retain branches of alternative interpretations. New independent measurements can add information, whereas reusing model predictions can perpetuate earlier errors. Because a local edit can change distant predictions, its effect includes both corrections near the target and changes in previously interpreted regions.

Local conditions cannot eliminate nonuniqueness in subsurface interpretation. Lateral facies changes, diagenesis, and unresolved heterogeneity can produce different properties in structurally similar regions, and incorrect conditions may yield continuous but inaccurate predictions. A limited set of blind wells cannot determine all interwell variation. Structural supervision here is learned regularization and does not directly impose geophysical forward-modeling constraints; adversarial training does not provide calibrated uncertainty.

4.4 Prospects for other indirect imaging applications

SHARAD interface extraction provides an example from another reflection-imaging modality, motivating investigation of local conditional prediction beyond seismic data. Similar problems occur in ultrasound contour identification and radar remote-sensing interpretation, where local contours or field surveys can supply information for target selection. Spatial relationships, errors, and valid conditions nevertheless differ across imaging systems.

5 Conclusion

GEM provides an effective conditional inference framework for geological interpretation when indirect observations do not fully determine the target. It encodes local interpretations and measurements as spatial conditions and combines them with image volumes in a shared three-dimensional model. Across surveys, the same weights and output head support structural interpretation, geobody identification, and property estimation, while revised conditions enable predictions to be updated without retraining. Field fault and channel comparisons, synthetic property evaluation, and blind-well predictions support the effectiveness of this approach across geometric, categorical, and continuous targets, with performance gains that depend on the setting. The radar application extends the demonstrated use of the framework to another reflection-imaging modality.

GEM treats interpretation and property modeling as inference from observations constrained by local knowledge. Conditions supply target identity, spatial extent, or measured values that the image alone cannot determine; learned spatial relationships guide their extension through the observed volume. This formulation also applies to indirect imaging problems in which domain knowledge or independent measurements can be spatially registered with observations, allowing predictions to be revised as those constraints change.

Data and code availability

Code website: `https://github.com/douyimin/GEM`. Pretraining and evaluation draw on public repositories and survey-specific data described in Appendix A.

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Supplementary Information

A Implementation and training details

A.1 Architecture and training data

The GEM training workflow (Supplementary Fig. S1) comprises masked backbone pretraining, independent training of a structure-aware network, and multitask joint training guided by local conditions.

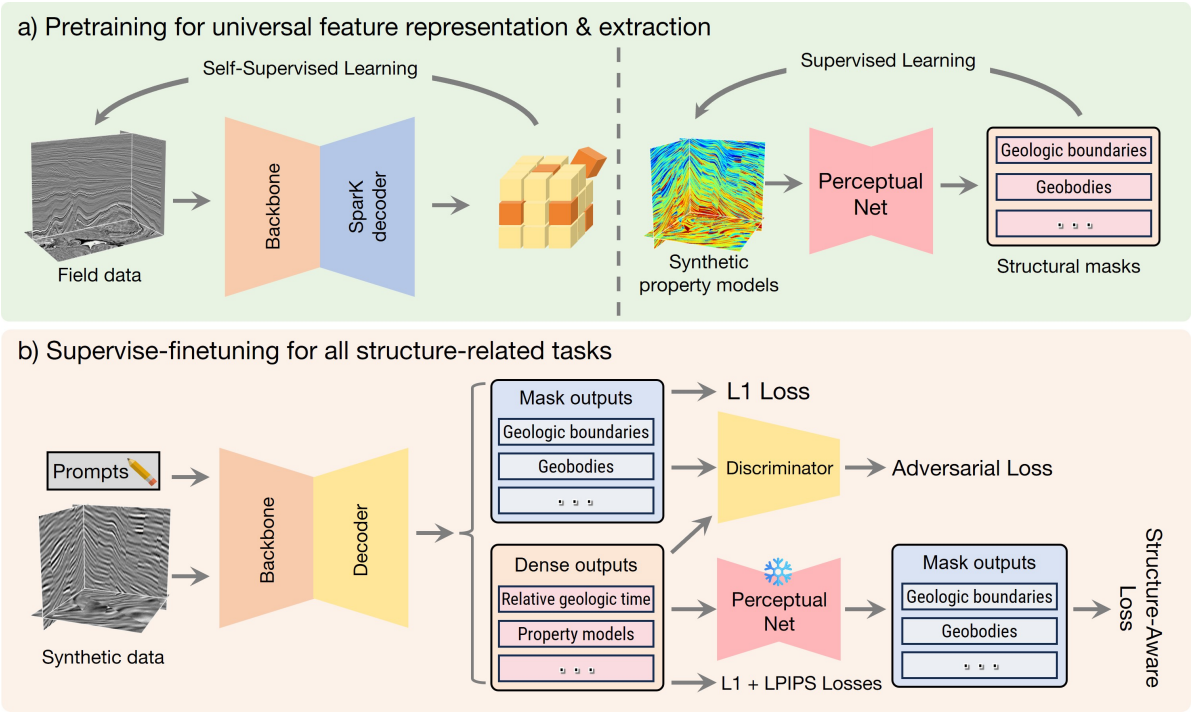

Figure S1. GEM training workflow. (a) Left: masked reconstruction pretraining of the seismic backbone. Right: independent training of the structure-aware network using synthetic property models and corresponding structural labels. (b) Observations and local conditions jointly drive multitask training, with targets including object masks, stratigraphic coordinates, and property fields. The structure-aware network remains frozen at this stage, providing structural supervision for dense outputs, together with voxelwise error, adversarial, and slice-based perceptual terms. Output groups represent different target types; the model shares its backbone and output head. Loss definitions and their scope are given in this appendix.

GEM's three-dimensional architecture, adapted from HRNet[26], contains approximately 670 million parameters. It retains quarter-resolution features and exchanges information with two lower-resolution branches, using a modified CBAM to recalibrate features[34]. Masked pretraining samples seismic blocks of $160 \times 160 \times 160$ voxels, masks 80% of the spatial input, and learns spatial dependencies between visible reflections and missing regions through reconstruction. The high masking ratio is intended to reduce redundancy between adjacent slices and encourage the use of broader spatial context. The convolutional architecture is shown in Supplementary Fig. S2. Synthetic training data contain spatially aligned images, physical properties, RGT, and structural labels corresponding to the same simulated volume. During training, local conditions are sampled from complete targets and supplied with images, while the complete targets supervise predictions (Supplementary Fig. S3).

Field training data include seismic data assembled for the SFM study[20] and public data repositories[35–38]. The training data include more than 500 field volumes. Synthetic data include 1,000 generated volumes of 512^3 voxels, supplemented by channel and cavity datasets[18,19].

A.2 Representation and encoding of local conditions

Condition encoding maps local information onto the observation grid and marks valid locations. Training conditions are sampled from complete labels; application conditions come from human interpretations, independent measure-

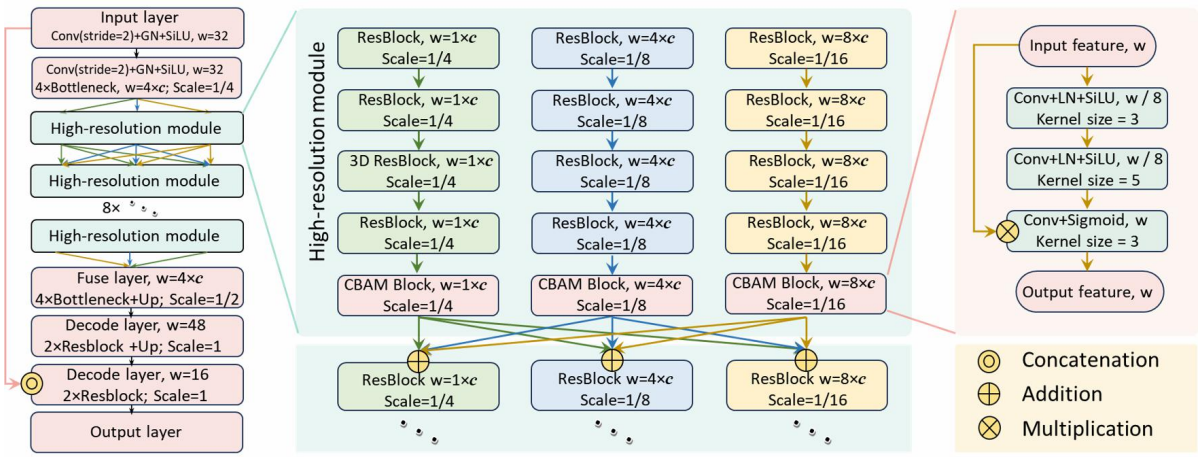

Figure S2. GEM backbone. In the three-dimensional convolutional architecture adapted from HRNet, a high-resolution branch exchanges features with lower-resolution branches, with modified convolutional attention modules.

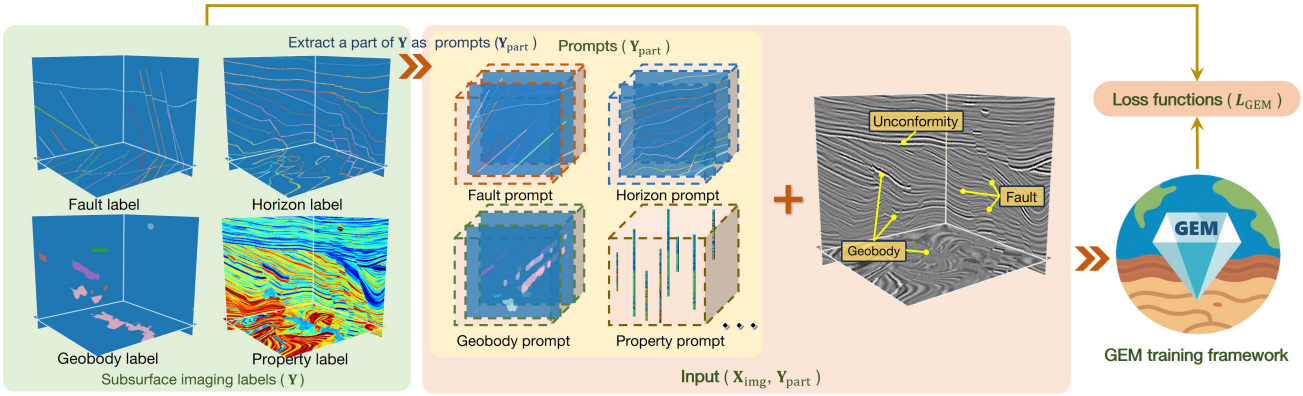

Figure S3. Training targets and local condition sampling. Left: fault, horizon, geobody, and property labels $\mathbf{Y}$. Local sections, interpretation lines, or well-like values are sampled from the current target to form conditions $\mathbf{Y}_{\text{part}}$, which enter GEM with subsurface images $\mathbf{X}_{\text{img}}$. Losses compare model predictions with the corresponding complete targets. Here $\mathbf{X}_{\text{img}}$ corresponds to $\mathbf{X}$ in the text, local conditions are encoded by condition values and validity masks, and L_{GEM} corresponds to Eq. (4).

Table S1. Training configurations.

Stage	Input dimensions / batch size	Optimizer / learning rate	Steps and notes
Masked pretraining	160^3 / 64	AdamW / 10^{-4}	500,000; masking ratio 0.8
Structural supervision network	Batch size: 96	AdamW / 10^{-3}	300,000; base width 24; subsequently frozen
Mixed-task training, stage 1	$320 \times 400 \times 128$ / 8	Adam / $10^{-4} \rightarrow 10^{-5}$ (G and D)	300,000; cosine annealing
Mixed-task training, stage 2	$512 \times 512 \times 128$ / 8	Adam / 10^{-5} (G and D)	200,000; fixed learning rate; activation checkpointing

ments, or reviewed predictions. Object interaction preserves the provenance and history of each condition, with merging rules described in Section A.4.

Object and geometric conditions. Interpretation lines or local regions are mapped to the inference grid and merged into a binary label volume $\mathbf{L}_j$, where j denotes the current object. In object interaction, the validity mask $\mathbf{M}_j$ is set to 1 over each entire annotated section and to 0 elsewhere. The specified foreground and the remaining background on that section therefore jointly form the input conditions; the actual condition extent is not restricted to visible lines. Conflicting labels at section intersections are handled during merging.

To represent geometric edges, the encoded values are

$$\mathbf{U}_j = \mathbf{M}_j \odot [2\,\mathrm{clip}(\mathbf{L}_j + 2(K_\sigma * \mathbf{L}_j), 0, 1) - 1]\,, \tag{2}$$

where K_σ is a three-dimensional Gaussian kernel with $\sigma = 1$ inference-grid sample, and smoothing precedes multiplication by the mask. The image, $\mathbf{U}_j$, and $\mathbf{M}_j$ form three input channels. Encoded values represent foreground and background, while the validity mask distinguishes annotated background from unlabeled space.

Continuous numerical conditions. Property conditions are supplied by well-log values registered to the image grid, and RGT conditions by relative coordinates on reference sections. Numerical processing must be consistent with training targets and output decoding, including units, normalization, valid intervals, and restoration of the prediction scale. Binarization and soft-edge transforms used for geometric conditions are not applied directly to continuous measurements. Continuous conditions and output scaling are described in Appendix A.5.

A.3 Joint learning from local information to three-dimensional targets

Learning complete targets from local conditions. Joint training samples target types and condition geometries. A target is selected from spatially aligned data, and sections, interpretation lines, or vertical profiles are sampled from it as conditions. Observations and conditions enter GEM, and complete targets provide supervision. The corresponding local values, valid extents, and encodings constitute conditional inputs for different targets. Complete labels serve both as a source for condition sampling and as a reference for prediction supervision, with both uses confined to training. Independent evaluation information should be kept separate from application conditions and model selection.

Joint training uses 1,000 spatially aligned synthetic volumes of 512^3 voxels and public channel and cavity datasets[18,19]. Structure, object, and continuous-field labels from the same simulated volume provide condition samples and prediction supervision for their respective targets. Data settings are given in Appendix A.

Initialization through masked pretraining. Before conditional joint training, the backbone uses masked image modeling adapted from SparK[21], reconstructing masked regions from visible seismic blocks to learn reflection organization and spatial context. Pretrained parameters initialize the prediction backbone, while target selection and conditional responses are learned during joint training. Pretraining uses more than 500 field seismic volumes together with synthetic inputs. Masking ratio, input dimensions, and training budgets are listed in Appendix A.

Learned structural supervision. A separate structure-aware network, F_{perc}, is trained to constrain geological spatial structure in dense predictions. The network takes seismic, property, and RGT fields as inputs and is supervised by structural labels such as faults, channels, and cavities; its parameters are subsequently frozen. For dense GEM predictions $\widehat{\mathbf{Y}}_d$ with corresponding structural references $\mathcal{S}$, we define

$$\mathcal{L}_{\mathrm{SAP}} = \mathcal{L}_{\mathrm{seg}}\big(F_{\mathrm{perc}}(\widehat{\mathbf{Y}}_d), \mathcal{S}\big), \tag{3}$$

where $\mathcal{L}_{\mathrm{seg}}$ combines Dice loss and cross-entropy, as specified in Appendix A.6. Gradients pass through the frozen perceptual network to GEM, constraining spatial structure in the predicted fields that can be assessed by structural recognition tasks. This network provides training supervision only and does not form a sequential prediction stage during application. The term is learned regularization and does not directly impose geophysical forward-modeling constraints.

Joint objective. The joint training objective is

$$\mathcal{L}_G = \alpha\mathcal{L}_1 + 0.01\mathcal{L}_{\mathrm{SAP}} + \mathcal{L}_{\mathrm{adv}} + \mathcal{L}_{\mathrm{LPIPS}}. \tag{4}$$

$\mathcal{L}_1$ constrains voxelwise deviations, $\mathcal{L}_{\mathrm{SAP}}$ constrains predicted fields with structural references, $\mathcal{L}_{\mathrm{adv}}$ regularizes appearance relative to the target distribution, and $\mathcal{L}_{\mathrm{LPIPS}}$ measures perceptual differences on slices along three orthogonal directions[39]. The weight α decreases from 30 to 2 through cosine annealing. Optimization details are given in Appendix A.

A.4 Interactive revision with fixed parameters

Each interaction round consists of prediction, review, and condition revision, followed by inference using the updated complete condition state (Supplementary Fig. S4). Model parameters remain fixed. The following describes object interaction for faults, horizons, and geobodies; continuous numerical conditions and RGT self-conditioning are handled separately.

Condition state and updates. Let $\mathcal{P}_j^{(k)}$ be the complete condition set for target j at round k, including human interpretations, revised prediction geometry, and inherited predicted foreground. Let $\mathcal{A}_j^{(k+1)}$ denote new or revised records and $\mathcal{R}_j^{(k+1)}$ denote old records that are withdrawn or replaced. Then

$$\begin{aligned} \mathcal{P}_j^{(k+1)} &= \big(\mathcal{P}_j^{(k)} \setminus \mathcal{R}_j^{(k+1)}\big) \cup \mathcal{A}_j^{(k+1)}, \\ \widetilde{\mathbf{Y}}_j^{(k+1)} &= G_\theta\big(\mathbf{X}_j^{(k+1)}, \mathcal{E}(\mathcal{P}_j^{(k+1)})\big). \end{aligned} \tag{5}$$

Here, $\mathcal{E}$ includes condition merging and the grid encoding in Section A.2; $\mathbf{X}_j^{(k+1)}$ is the resampled and normalized image within the current inference region; and $\widetilde{\mathbf{Y}}$ is the prediction on the inference grid. Results are subsequently mapped back to the original grid. Each round uses all valid conditions for the target, not only the most recent edit.

Addition, replacement, and withdrawal. Interpretation lines or local foreground regions are added for omitted objects or incomplete areas. For geometric errors, thresholded predictions on a selected section are converted into editable polylines. Their position, orientation, and extent are adjusted and saved as conditions for the target. The revised geometry is retained when selected again. Candidate results from specialized models can enter the workflow through the same conversion. Old conditions can be replaced or withdrawn to correct earlier judgments.

For incorrectly connected branches or non-target regions, users specify exclusion regions. On submission, the object's section foreground at the current threshold is saved. The next round first merges human positive conditions with these frozen foregrounds and then subtracts all exclusion regions, which take precedence wherever they overlap. Frozen foregrounds do not change with subsequent threshold adjustments or predictions and are retained, replaced, or withdrawn with their associated records. They inherit information, and potentially errors, from candidate interpretations, so their predictive provenance is always retained.

History preservation and three-dimensional response. Each update saves the complete condition state and parent round, enabling backtracking and branches of alternative interpretations. History enters the next input through explicit records rather than through accumulation in an internal network state across rounds. Contradictory human labels at section intersections are resolved before encoding. Intersections involving inherited foreground follow the foreground-merging and exclusion-priority rules. Geometric extraction and record-restoration details are provided in Appendix A.8.

Conditions are assembled and inference is performed independently for each target. The inference region may be fixed or determined from the joint extent of positive prompts and frozen foreground. Targets that are not updated retain their existing results, whereas predictions for the edited target may change outside the edited region. A local edit therefore triggers three-dimensional prediction under revised conditions.

A.5 Continuous conditions and output scaling

Continuous conditions retain their measurement units and valid intervals and are registered to the seismic grid. Predictions are restored to the physical scale using the normalization parameters of the input well logs. Lithology uses categorical encoding.

A.6 Training objective of the structural supervision network

The independent structure-aware network uses the same architecture family and is trained on complete structural labels. Its segmentation objective combines Dice loss and cross-entropy:

$$\begin{aligned} \mathcal{L}_{\mathrm{seg}}(p,t) = \frac{1}{C}\sum_{c=1}^{C}\left(1 - \frac{2\sum_i p_i^{(c)} t_i^{(c)} + \epsilon}{\sum_i p_i^{(c)} + \sum_i t_i^{(c)} + \epsilon}\right) \\ - \frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} t_i^{(c)} \log p_i^{(c)}. \end{aligned} \tag{6}$$

Here N is the number of voxels, C the number of classes, $p_i^{(c)}$ and $t_i^{(c)}$ the predicted probability and reference label, and ϵ a numerical stabilization term. After independent training, network parameters are frozen, and structural

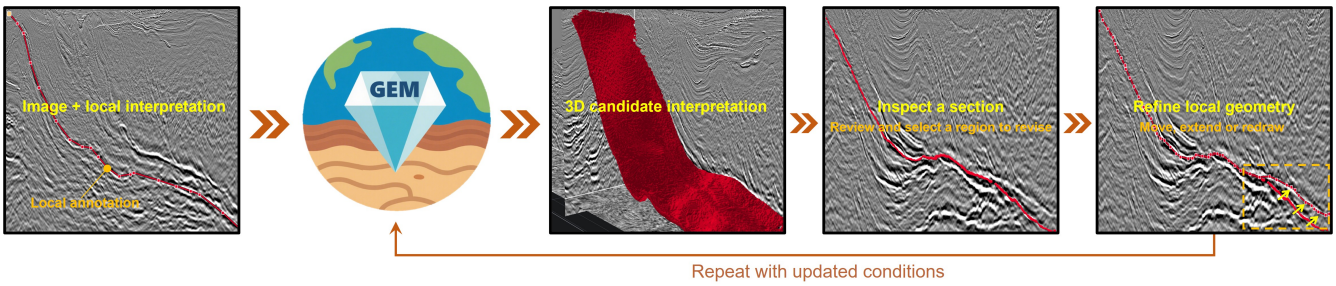

Figure S4. Interactive revision with fixed parameters. Seismic images and local interpretations jointly enter GEM to produce a three-dimensional candidate interpretation. Interpreters inspect intersections of predicted surfaces with sections, move, extend, or redraw local geometry in selected regions, and use the revisions as updated conditions for the next inference round. Red surfaces and curves denote candidate interpretations and their section intersections; nodes on the curves enable local editing. Yellow dashed boxes and arrows mark revision regions. Model parameters remain fixed across rounds, and valid historical conditions are retained.

supervision is supplied to GEM's dense outputs through Eq. (3). Freezing parameters still allows gradients to pass through the network to the predictions.

A.7 Optimization settings and adversarial objective

Input dimensions, batch sizes, learning rates, and iteration counts for each stage are listed in Supplementary Table S1. Masked pretraining uses AdamW with a learning rate of 10^{-4} and $(\beta_1, \beta_2) = (0.9, 0.999)$. The structural supervision network uses AdamW and a base channel width of 24. Mixed-task training uses Adam with $(\beta_1, \beta_2) = (0, 0.9)$. In stage 1, both the generator and discriminator start at 10^{-4} and decrease to 10^{-5} through cosine annealing over 300,000 steps. In stage 2, both continue for 200,000 steps at a fixed learning rate of 10^{-5}. Inline and crossline directions are randomly exchanged during training. Stage 2 increases the input volume and enables activation checkpointing.

The adversarial term uses the relativistic average least-squares formulation[40]. Let $C_r = C(\mathbf{Y}_r)$ and $C_f = C(\widehat{\mathbf{Y}})$ denote discriminator scores, and $\bar{C}_r$ and $\bar{C}_f$ their corresponding batch expectations. Then

$$\mathcal{L}_D = \mathbb{E}[(C_r - \bar{C}_f - 1)^2] + \mathbb{E}[(C_f - \bar{C}_r + 1)^2], \tag{7}$$

$$\mathcal{L}_{\text{adv}} = \mathbb{E}[(C_f - \bar{C}_r - 1)^2] + \mathbb{E}[(C_r - \bar{C}_f + 1)^2]. \tag{8}$$

This objective constrains the statistical appearance of outputs relative to training targets. Discriminator scores do not represent geological validity or posterior probabilities[41].

A.8 Construction of geometric conditions

The object-interaction workflow is shown in Supplementary Fig. S4, and condition updates are described in Section A.4. Predicted section intersections are first reduced to centerline backbones from connected components and then simplified into polylines that retain major turning points. Short local gaps are connected according to endpoint distance, orientation, and the presence of other predicted horizons between them, reducing erroneous connections across layers. When an exclusion region is moved, locations no longer covered can be restored from the foreground stored with that record. Withdrawing a record also withdraws its associated foreground. Complete condition records preserve both visible lines and their associated foreground.

A.9 Computational cost

All three stages use eight NVIDIA H20 GPUs, each with 96 GB of memory. Pretraining takes approximately 1.8 seconds per iteration, or about 10 days for 500,000 iterations. The two mixed-task training stages take approximately 1.7 and 3.2 seconds per iteration, respectively, for about 13 days in total. These times exclude independent training of the structural supervision network.

For an inference volume of $512 \times 512 \times 128$ voxels, the computational cost is 5.4×10^4 GFLOPs and float16 memory use is approximately 30 GB. These estimates apply to the stated configuration.

Interaction can use reduced-resolution GPU inference, and some full-resolution dense outputs can be computed on CPUs with larger memory. Resampling and possible tiled stitching affect accuracy and latency.

A.10 RGT self-conditioning

RGT self-conditioning initializes prediction with a central section whose values range from 0 to 1. Four additional sections are then selected from the predicted field as conditions to recompute RGT (Supplementary Fig. S5). The process can be iterated.

The additional sections come from model predictions and provide no independent measurements. Scaling the reference section defines only a relative stratigraphic coordinate.

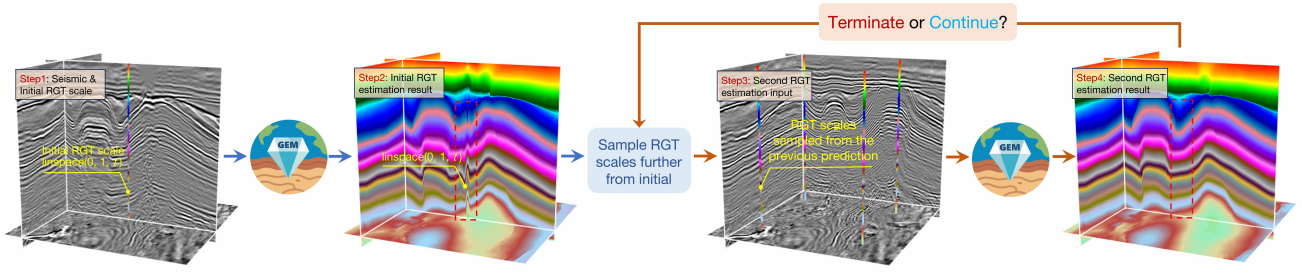

Figure S5. RGT self-conditioning. A reference section initializes prediction, and additional sections from the predicted field provide conditions for subsequent computations.

B Supplementary qualitative results

Supplementary Fig. S6 compares fault results from Hikurangi and Baiyun, showing responses of different methods to complex surfaces and deep structures. Supplementary Fig. S7 provides channel and cavity comparisons, with SAM 2 extended through a slice/video-based workflow. Supplementary Fig. S8 shows property comparisons for Teapot and F3, and sonic transit time and natural gamma-ray predictions around complex structures at Poseidon. These comparisons use the existing adaptation workflow of each method; the scope of quantitative interpretation is specified in Appendix C.

C Evaluation data and scope of result interpretation

C.1 Evaluation protocol

GEM applications are reported as fixed-weight conditional inference: parameters are not updated on application data, and target information is supplied through local conditions. Transfer across surveys, geological categories, properties, and task families requires holding out data regions, object categories, target physical quantities, and prediction operations, respectively; none follows from fixed weights alone. Some baselines use the corresponding conditions as weak training labels. Results therefore compare their respective adaptation workflows without individually controlling all differences in pretraining, capacity, and optimization. Object tasks use IoU, synthetic properties use full-field evaluation, and field properties use blind-well comparisons.

C.2 Evaluation data and conditioning

Synthetic fault testing includes 100 volumes of 512^3 voxels, and field fault evaluation uses 20 expert-annotated slices from Hikurangi and Baiyun. Channel evaluation includes 100 synthetic samples and 30 annotated field slices from Parihaka and Romney. Synthetic cavity evaluation uses 20 samples. Field slices are spatially correlated within surveys and cannot be treated as independent survey samples. Reported IoU values describe segmentation overlap within the annotated regions.

Conditions and outputs for each task are listed in Supplementary Table S2. In object interaction, a complete section mask may contain many more conditioned voxels than a single interpretation line and may include background constraints and inherited predictions. Reported prompt counts describe instance workflows and are not standardized to equal human time or equal numbers of annotated voxels. Model-generated foreground is not counted as independent human measurement.

SEAM I predictions are evaluated against the complete synthetic velocity field. GEM uses four well-like sections as conditions, while baselines are trained on the same sections. Field property evaluation uses blind wells outside the conditioning set: F3 uses three conditioning wells and the F02-1 blind well; Teapot metrics aggregate four evaluation wells, two of whose curves appear in the main figure; and Delft shows comparisons for PNA-13 and PNA-GT-01.

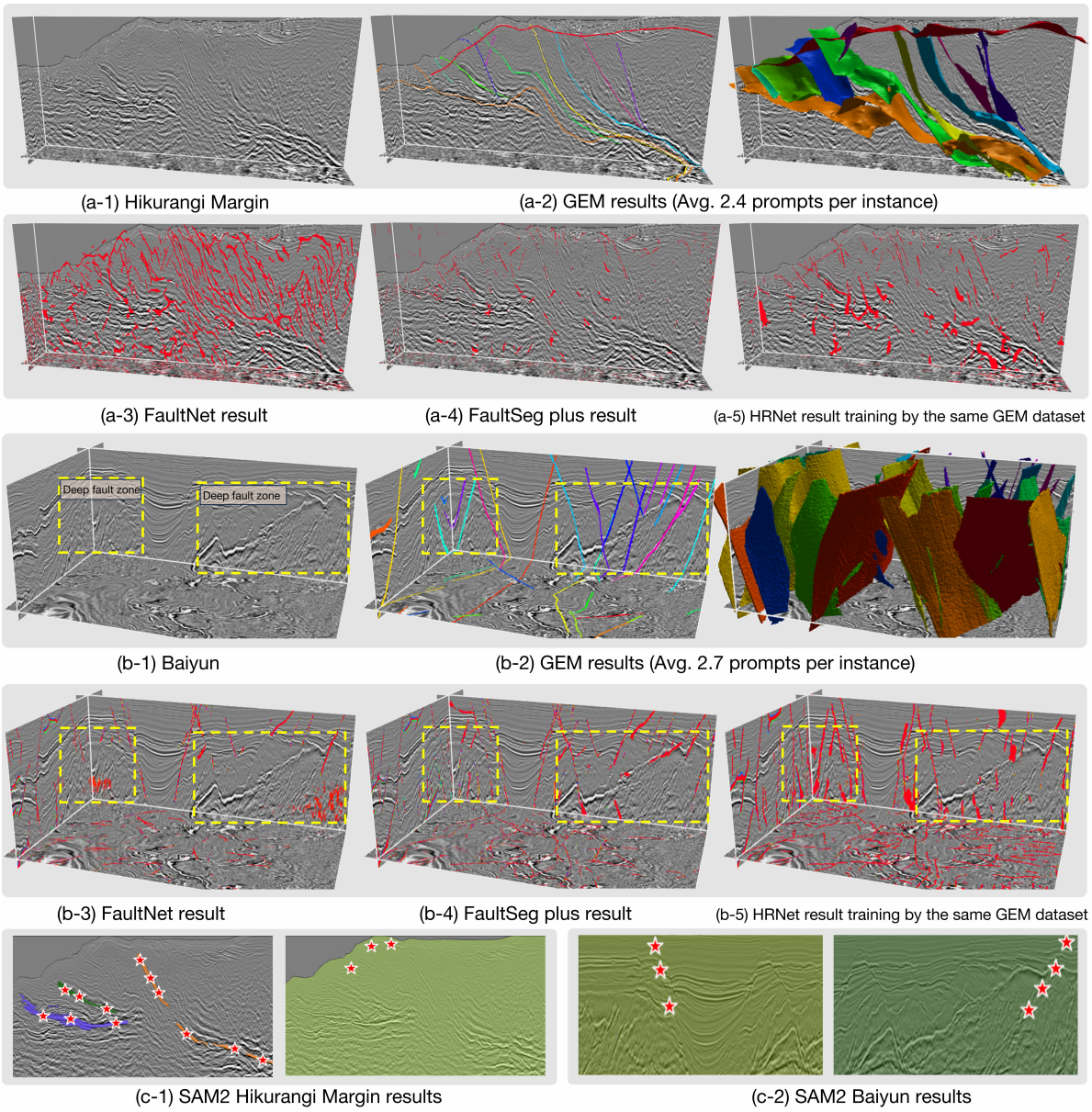

Figure S6. Supplementary fault comparisons for Hikurangi and Baiyun. Each group shows seismic sections and predictions from different methods. GEM receives local target conditions. Comparisons reflect differences in target coverage and geometry under the corresponding adaptation workflows.

Table S2. Conditions and outputs for subsurface imaging tasks.

Target	Local conditions	Output representation
Object identification	Membership and exclusion regions	Object mask
Interface interpretation	Interpretation lines and local interface positions	Three-dimensional interface representation
Stratigraphic coordinates	Reference sections or relative coordinates	RGT field
Property modeling	Registered local measurements	Continuous numerical field

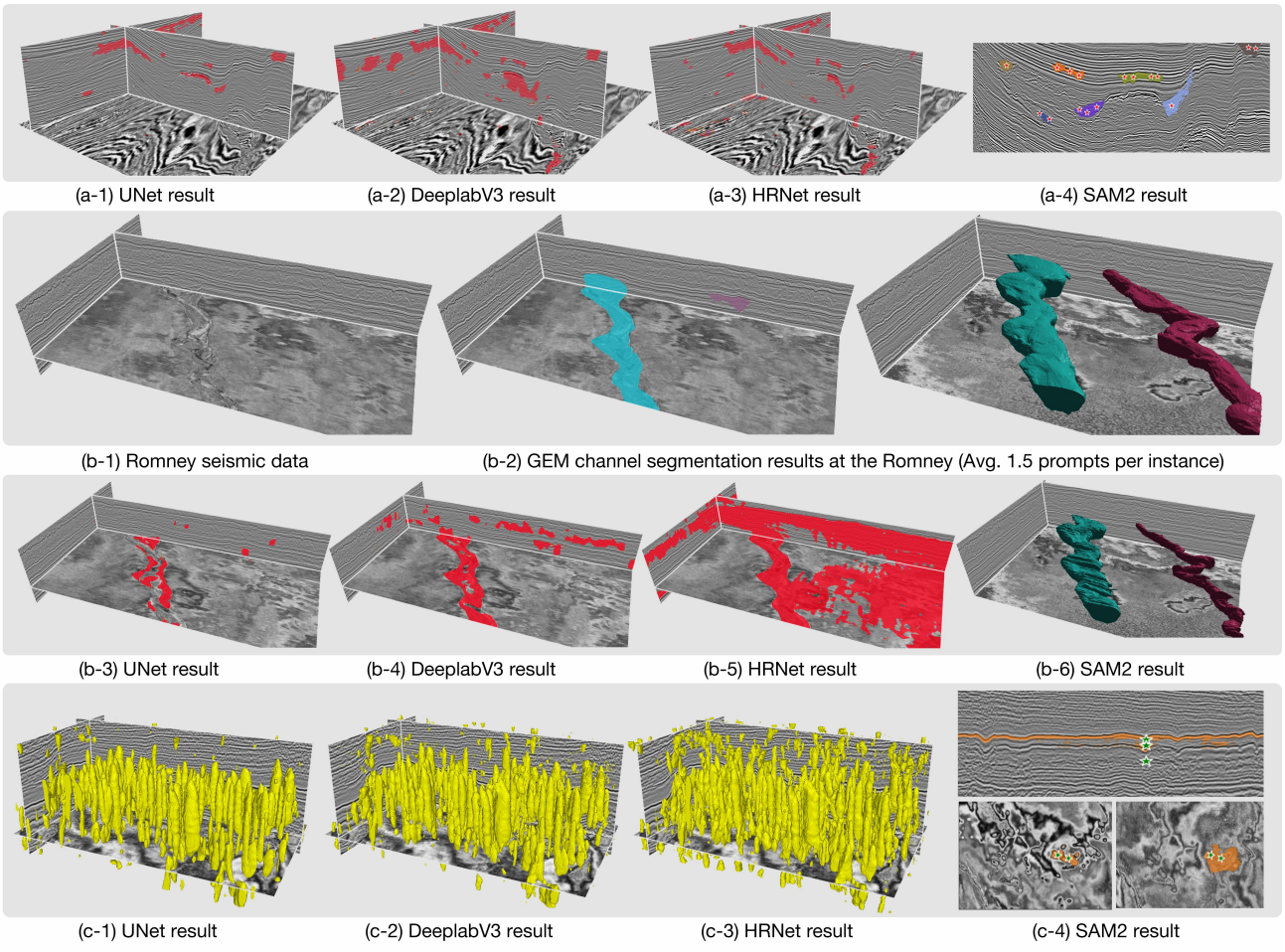

Figure S7. Supplementary channel and cavity comparisons. SAM 2 channel predictions use a slice/video-based extension, illustrating differences in continuity across slices. Synthetic cavity predictions and reference labels are also shown.

Metrics retain their original scales, without comparing MAE across datasets. Lithology is categorical and is not evaluated by the MAE used for continuous properties.

GEM weights remain fixed during application. F3 natural gamma-ray and lithology results demonstrate heterogeneous-target applications.

The SHARAD example illustrates radar reflection-interface extraction. Relevant sources cite three-dimensional imaging studies of midlatitude and polar regions[30,31], respectively. Its results are used only to assess interface extraction.

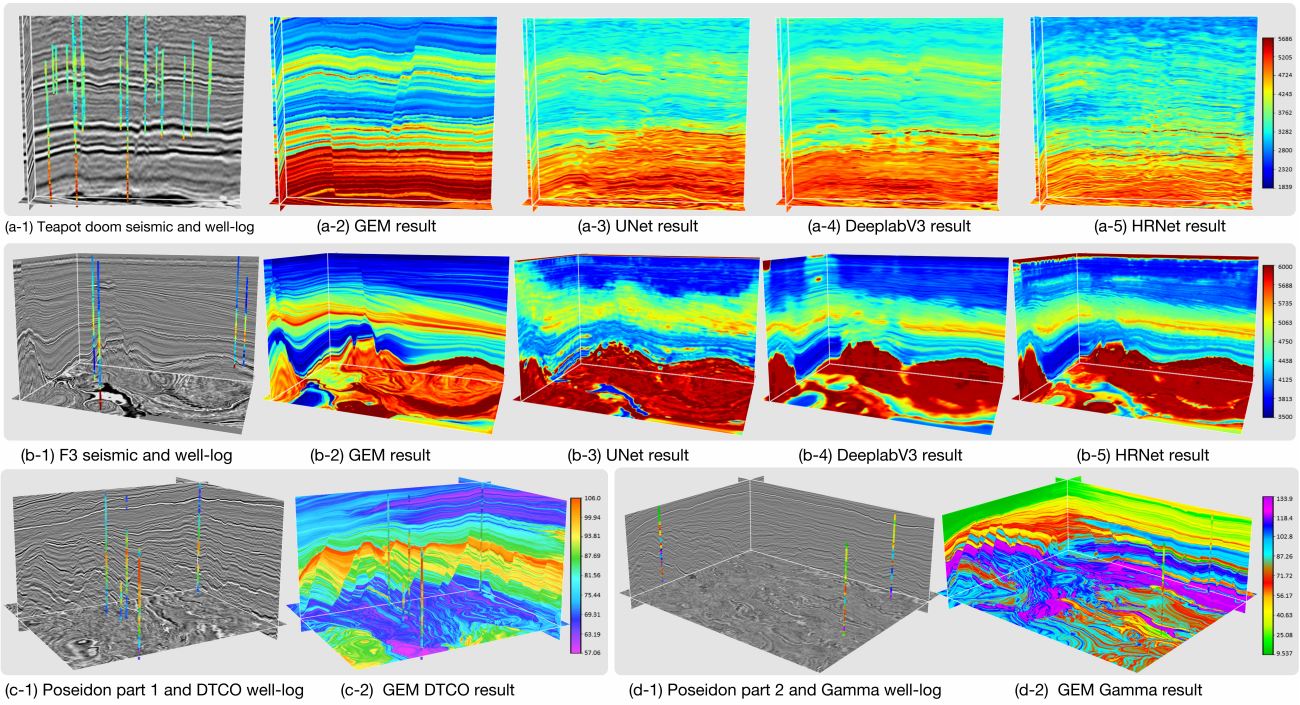

Figure S8. Supplementary property-modeling examples. Teapot and F3 show acoustic impedance predictions from GEM and learning-based baselines. Poseidon shows sonic transit time (DTCO) and natural gamma-ray fields in a setting with complex faults and unconformities. Property predictions are juxtaposed with seismic sections to compare property boundaries and reflection structures.